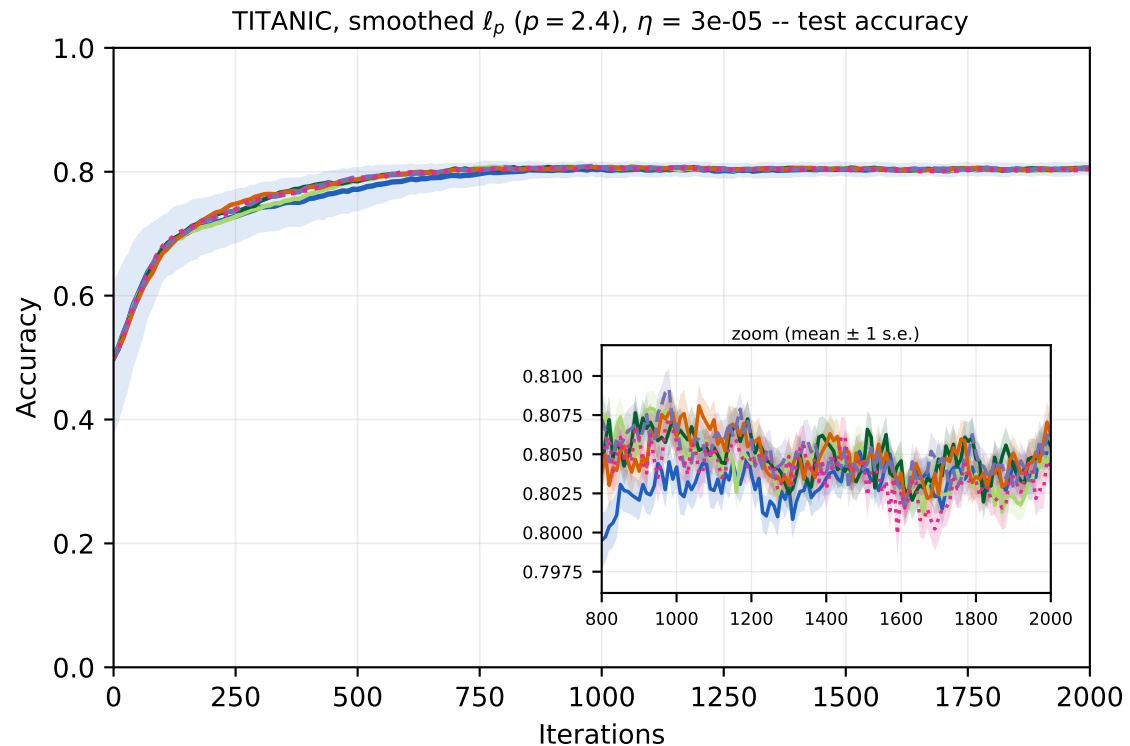
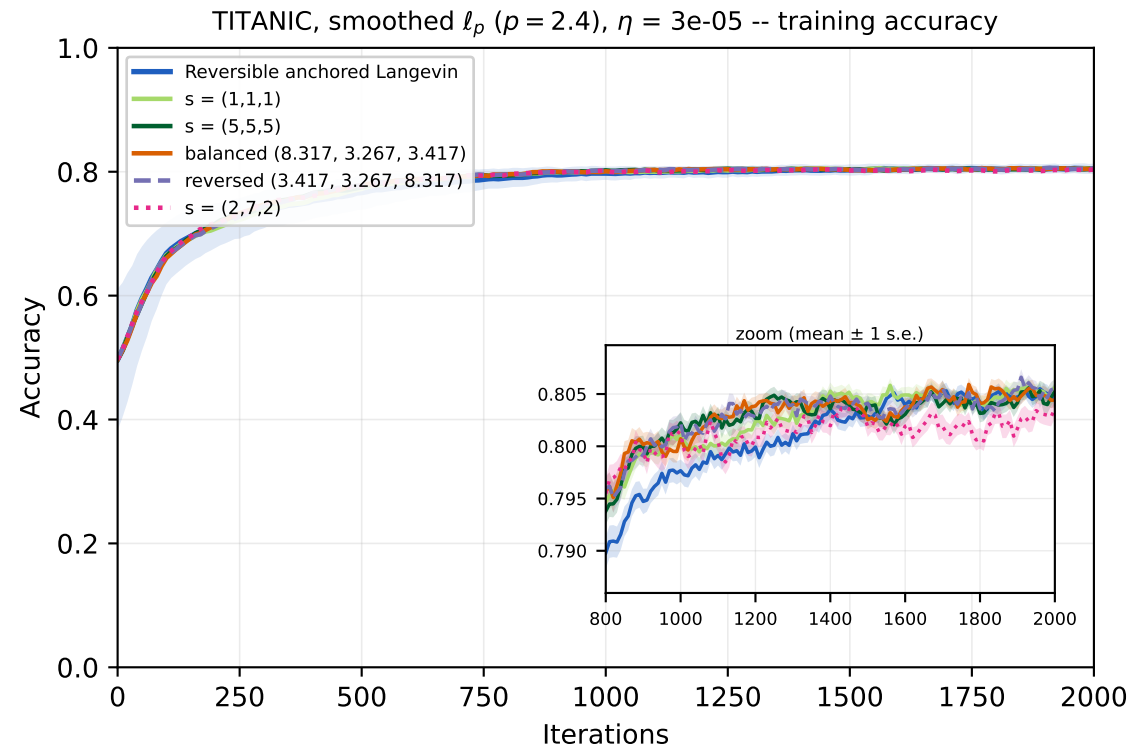


titanic_lp: per-block strengths at $\eta = 3\text{e-}05$ (mix)



TITANIC -- smoothed ℓ_p $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{(p/2)} \leq \text{Lambda}$, $p = 2.4$, $\text{eps} = 0.18$, $\text{Lambda} = 4$. $d = 10$ (intercept w_0 + 9 features). $U = \sum \text{softplus} - y x.w + w_0^2 / (2 \text{sigma}^2) + \text{lambda} \sum_{\{j \geq 1\}} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \text{delta}^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\text{sigma} = 10$, $\text{lambda}_{\text{lasso}} = 7.12 = 0.01 n_{\text{train}}$, $\text{delta} = 0.0108$ (a in $[0.50, 1]$). Step $\text{eta} = 3\text{e-}05$. Per-block strengths $s = (s_1, s_2, s_3)$ on $l_1 = \text{Age/SibSp/Parch}$, $l_2 = \text{Fare/is_female/Pclass}_2$, $l_3 = \text{Pclass}_3/\text{Embarked}_Q/\text{Embarked}_S$. Isotropic vs gradient-balanced vs reversed-balanced vs hand-picked. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Band: mean ± 1 SD (ddof = 1) for the reversible arm -- repeat-run variability at a fixed split, not a confidence or credible interval. All triples are reported in results/blocks/blocks.csv; nothing here was selected.